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I Semester M.C.A. (2 Years Course) Examination, August/September 2021
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA4 : Theory of Computation

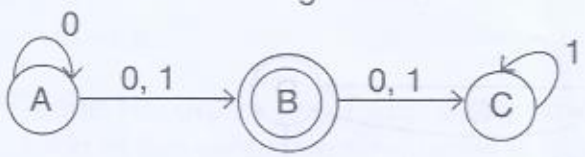
Time : 3 Hours Max. Marks : 70

Instruction : *Part – A : Answer **any five** questions.*
*Part – B : Answer **any four** questions.*

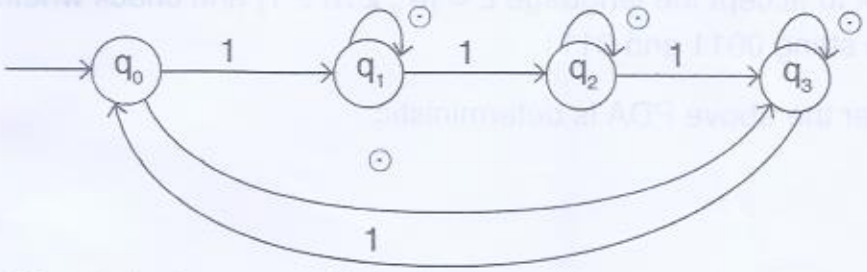
PART – A

Answer **any five full** questions. **(5×6=30)**

1. Define DFA, NFA and E-NFA with examples.
2. Convert the following NFA to DFA.



3. State and prove pumping lemma for regular languages and show that L = {ww/w belongs to {a, b}*}.
4. Write the left most and right most derivation for the following production and check whether it is ambiguous.
   $S \rightarrow iCtS/iCtSeS/a$
   $C \rightarrow b$.
5. Explain with examples Chomsky hierarchy of languages.
6. Convert the following Moore Machine to an equivalent Mealy Machine.



Given $\lambda(q_0) = a$, $\lambda(q_1) = b$
$\lambda(q_2) = c$, $\lambda(q_3) = d$.

P.T.O.

7. Explain Turing Machine and instantaneous description of Turing machine.

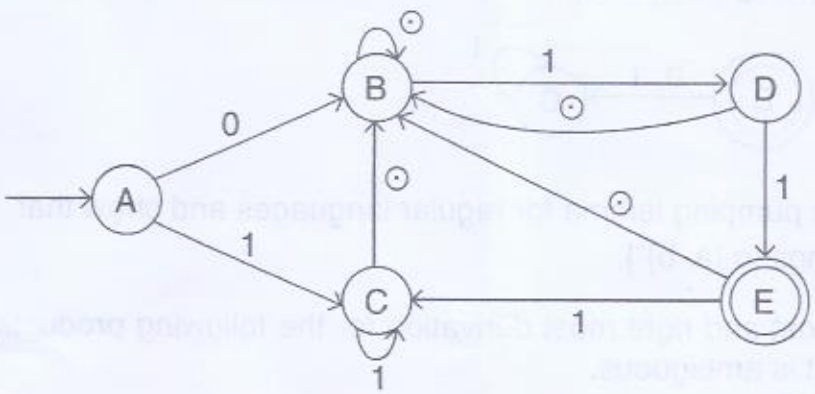
8. Define post correspondence problem. Check whether the lists
   $A = \{b, bab^3, ba\}$ and $B = \{b^3, ba, ba\}$
   have a PCP solution.

PART – B

Answer **any four full** questions. **(4×10=40)**

9. a) Design a DFA to accept strings of a's and b's starting with atleast two a's and ending with atleast two b's. 4

   b) Minimize the following DFA. 6



10. a) Construct an E-NFA for the regular expression $0(0 + 1)^* 01$. 5

    b) Prove that context five languages are closed under union, concatenation and star. 5

11. a) Design a PDA to accept the language $L = \{0^n, 2^n/n \geq 1\}$ and check whether it accepts the string 0011 and 0111. 8

    b) Check whether the above PDA is deterministic. 2

12. a) Convert the following grammar into equivalent CNF. 6

S → OA/IB

A → OAA/IS/I

B → IBB/OS/O.

b) Define GNF and write the steps to convert CFG to GNF. 4

13. a) Write short notes on Rice theorem and Halting problem. 6

b) Obtain a PDA for the CFG given below. 4

S → aABB/aAA

A → aBB/a

B → bBB/A

C → a.

14. Define Recursive and Recursively enumerable languages and prove that the union of two recursive languages is recursive. 10

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I Semester M.C.A. (2 Years Course) Examination, July 2022
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
1MCA4 : Theory of Computation

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any 5** questions.*
*2) Part – **B** : Answer **any 4** questions.*

PART – A

Answer **any five full** questions. **(5×6=30)**

1. Explain DFA and NFA with suitable examples.

2. State and prove pumping lemma for regular languages and prove that the following language is not regular.
   $L = \{\omega\omega | \omega \in \{a, b\}^*\}$

3. Minimize the following DFA :

δ	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_1	q_3
q_2	q_1	q_2
q_3	q_1	q_4
$* q_4$	q_1	q_2

4. Explain with example Chomsky hierarchy of generative grammar.

5. Construct a turing machine for the following language.
   $L = \{a^n b^n / n \geq 1\}$.

6. Explain Moore and Mealy machines.

P.T.O.

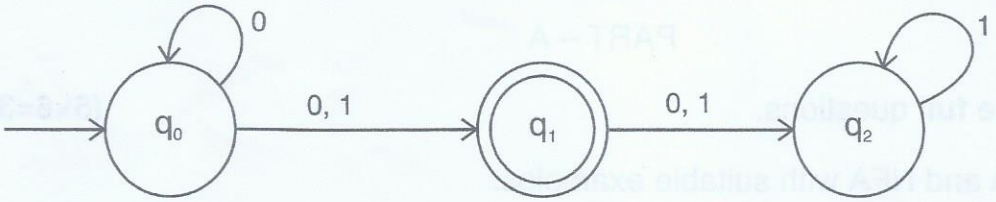
7. Prove that the complement of a recursive language is recursive and the union of two recursive languages is recursive.

8. Explain primitive recursive functions and μ-recursive functions.

PART – B

Answer **any four full** questions : **(4×10=40)**

9. a) Convert the following NFA into its equivalent DFA. **6**



b) Write short notes on the applications of finite automata. **4**

10. a) Design a DFA to accept the set of all strings not containing the substring 00 for $\Sigma = \{0, 1\}$ and show the acceptability of the string 101011. **6**

b) Explain $\in$-closure and write the steps to find out $\in$-closure. **4**

11. a) Check whether the following grammar is ambiguous. **5**

$S \rightarrow iCtS/iCtSeS/a$

$C \rightarrow b$

b) Define push down automata and language accepted by a PDA. **5**

12. a) Convert the following grammar into CNF. **6**

$S \rightarrow bB/aA$

$A \rightarrow aBB/aS/bA/a$

$B \rightarrow aBB/bS/b$

b) Explain GNF with example. **4**

13. a) Explain turing machines that accepts empty language. **5**

b) Define post correspondence problem with an example. **5**

14. Write short notes on :

a) Halting problem. **5**

b) Rice's theorem. **5**

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I Semester M.C.A. (Two Years Course) Examination, July 2023
(CBCS) (2020-2021 and Onwards)
COMPUTER SCIENCE
1 MCA 4 : Theory of Computation

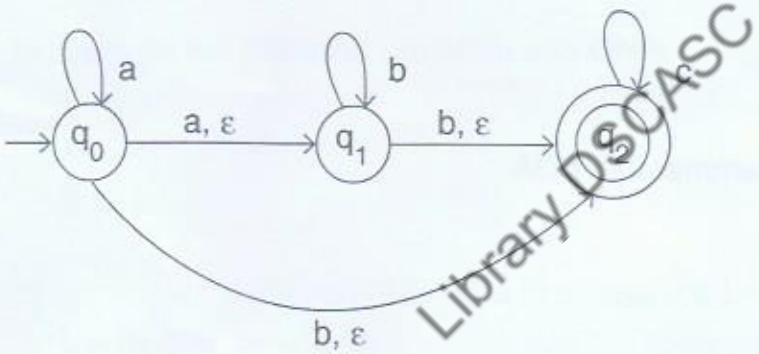
Time : 3 Hours | Max. Marks : 70

Instructions : *Part – **A** : Attend **any five** questions.*
*Part – **B** : Attend **any four** questions.*

PART – A

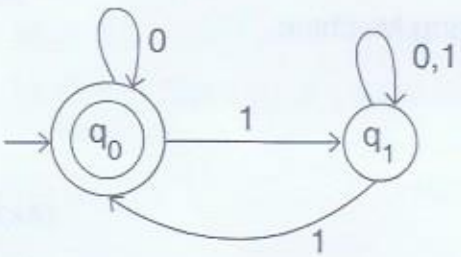
Answer **any five full** questions. **(5×6=30)**

1. Define ε-closure. Find ε-closure of all states for the given ε-NFA.



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2. Convert the following NFA to DFA.

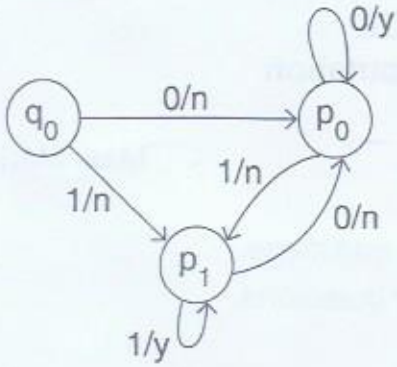


3. Regular languages are closed under union and intersection. Justify.

4. State and prove pumping lemma for regular languages.

P.T.O.

5. Convert the following Mealy machine to Moore machine.



6. Define ambiguous grammar. Check whether the following grammar is ambiguous.

   $S \rightarrow aB \mid bA$

   $A \rightarrow aS \mid bAA \mid a$

   $B \rightarrow bS \mid aBB \mid b$

7. Convert the following grammar into PDA.

   $S \rightarrow aABC$

   $A \rightarrow aB \mid a$

   $B \rightarrow bA \mid b$

   $C \rightarrow a$

8. Write a short note on Halting problem of Turing Machine.

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PART – B

Answer **any four full** questions. **(4×10=40)**

9. a) Explain extended transition function of DFA. **4**

   b) Draw a DFA to accept strings of a's and b's having even number of a's and even number of b's. **6**

10. Define equivalent and non equivalent states and minimize the following DFA. **10**

		1	0
→	A	B	C
	B	C	D
	C	C	E
*	D	D	D
*	E	E	E
	F	F	E

11. a) Write a short note on Chomsky Hierarchy of languages. **5**

    b) Construct an NFA with ε-moves for Regular Expression $(a+b)^*$ cd. **5**

12. a) Obtain a CFG for the language
    $L = \{a^n b^n \mid n \geq 1\}$. **4**

    b) Convert the following grammar into GNF. **6**

    $S \rightarrow A$

    $A \rightarrow aBa|a$

    $B \rightarrow bAb|b$

13. a) Design a Pushdown Automata to accept the Language
    $L = \{w \subset w^R | w \in (a+b)^*\}$ **8**

    b) Check whether the string a a b c b a a is accepted or not. **2**

14. a) Write a short note on variations of Turing Machine. **5**

    b) Prove that if a Language L and its complement L' both are RE then L is recursive. **5**

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I Semester M.C.A. (Two Years Course) Examination, June/July 2024
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1 MCA 4 : Theory of Computation

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Section – **A** and **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define a regular expression. Write a regular expression for the set of strings having alternation of 0s and 1s.

2. Design NFA to accept set of all strings whose third symbol from the left end is b. Find the complement automaton.

3. Prove that regular languages are closed under union, intersection and Kleene closure. Write a regular expression for even number of a's and even number of b's of a string w = {a, b}*.

4. Design a Moore Machine to determine residue mod 3 of input decimal number treated as binary string.

5. Explain Halting Problem of a Turing Machine. Is it solvable or unsolvable problem ? Discuss.

6. Convert the following grammar to Chomsky Normal form.

   $S \rightarrow A|AB0|A1A$

   $A \rightarrow A0|\in$

   $B \rightarrow B1|BC$

   $C \rightarrow CB|CA|1B.$

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P.T.O.

7. Consider the given ε-NFA. Convert to DFA.

δ_N	ε	a	b	c
→p	Φ	{p}	{q}	{r}
q	{p}	{q}	{r}	Φ
*r	{q}	{r}	Φ	{p}

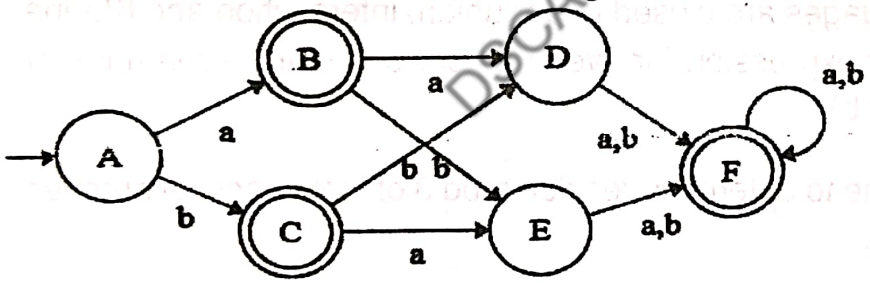
8. Explain the terms with an example each.
   a) e-closure of a state
   b) Null and unit productions
   c) Universal Turing Machine.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. Construct NFA that accepts set of all strings that end in 01. Give the extended transition function for the input string 00101. Also construct DFA for the above NFA using subset construction method.

10. When are two states said to be distinguished ? Minimize the given DFA.



11. a) State and prove pumping lemma for context free languages. Show that L= $\{a^n b^n c^n / n>=1\}$ is not context free.
    b) Explain the significance of the Chomsky's hierarchy for languages.

12. a) Define the pushdown automata based on the types of acceptance.
    b) Design pushdown automaton to accept all palindromes of odd length by empty stack.

13. Convert to Greibach Normal form {S → AB, A → SA|AA|a, B → SB|b}.

14. Construct a Turing Machine for proper subtraction, which is defined as m – n if m > n and 0 otherwise.

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I Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1 MCA 4 : Theory of Computation

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions.*
*2) Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five full** questions. **(5×6=30)**

1. What are the central concepts of automata theory ?
2. Obtain a NFA to accept the following language
   $L = \{ w/w \in abab^n \text{ or } aba^n \text{ where } n \geq 0\}$.
3. State and prove pumping lemma for regular languages and show that
   $L = \{a^{k^2} \mid K \text{ is an integer, } K \geq 1\}$ is not regular.
4. What is left recursion ? Eliminate left recursion from the following grammar
   $E \rightarrow E + T|T$
   $T \rightarrow T * F|F$
   $F \rightarrow (E)|$ id.
5. Discuss in detail the types of grammars.
6. Eliminate all unit productions from the grammar.
   $S \rightarrow AB$
   $A \rightarrow a$
   $B \rightarrow C|b$
   $C \rightarrow D$
   $D \rightarrow E|bc$
   $E \rightarrow d|Ab$.
7. Convert the following grammar into GNF
   $S \rightarrow AB$
   $A \rightarrow BSB$
   $A \rightarrow a$
   $B \rightarrow b$.

P.T.O.

8. Write a short note on :

   a) Halting problem of Turing machine. **3**

   b) Post correspondence problem. **3**

PART – B

Answer **any four full** questions. **(4×10=40)**

9. Consider the following $\in$ – NFA

δ	$\in$	a	b
$\rightarrow$ p	{ r }	{ q }	{ p, r }
q	ϕ	{ p }	ϕ
* r	{ p, q }	{ r }	{ p }

   a) Compute the $\in$ – closure of each state. **3**

   b) Convert the automation to a DFA. **7**

10. Explain closure properties of regular languages.

11. Check whether the following grammar is ambiguous
    S $\rightarrow$ AB|C
    A $\rightarrow$ aAb|ab
    B $\rightarrow$ cBd|cd
    C $\rightarrow$ aCd|aDd
    D $\rightarrow$ bDc|bc.

12. Construct a PDA to accept a string of balanced parentheses. The parentheses to be considered are (,), [,].

13. Construct a Turing machine to accept the language L = { $a^n b^n$| $n \geq 1$}.

14. Write a short notes on :

    a) Moore and Mealy machine. **6**

    b) Rice theorem. **4**

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